

Chemistry Ch 4.7: The Halogens (Group 17)

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Pehchan

Bairooni configuration $ns^2 np^5$, yani **7 bairooni electrons**. Sirf **ek** ki kami hai, is liye ye **bohat reactive** hain aur **-1** ion banate hain.

Shakal: F_2 halka peela gas · Cl_2 sabzi maael peela gas · Br_2 surkhi maael bhoora **maaye** · I_2 bhoori siyah **thos**.

Neeche jate hue **density barhti** hai aur **reactivity ghatti** hai.

Do tarteebein jo yaad karni hain

Reactivity aur displacement: $F_2 > Cl_2 > Br_2 > I_2$

Yani oopar wala neeche wale ko us ke namak se **nikal** deta hai. Misal: $Cl_2 + 2KBr \rightarrow 2KCl + Br_2$. Ulta nahin hoga.

Hydrogen halides ki garmi se mazbooti: $HF > HCl > HBr > HI$

Yani HF sab se mazboot hai aur HI sab se jaldi toot jata hai.

Dono tarteebein ek jaisi hain

Dono mein **F sab se oopar aur I sab se neeche**. Wajah wohi purani: F ka atom **sab se chhota** hai, to wo electron ko sab se zyada zor se kheenchta hai. Isi se reactivity bhi zyada aur bond bhi mazboot.

Quiz

1. Give the appearance and physical state of the four halogens, and say how density and reactivity change down the group.
2. Write the displacement order and the thermal stability order of the hydrogen halides. Will Br_2 displace Cl_2 from KCl? Explain.